Maxwell's Equations in Another Language
Wedge Products, Exterior Derivatives, and Hodge Duality

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Maxwell's equations and Einstein's field equation are usually presented in very different mathematical languages: the former as a set of coupled vector partial differential equations, the latter as a tensor equation on a curved manifold. In this essay we show that, once electrodynamics is rewritten in the language of differential forms, the two theories share a common grammatical structure: a connection, a curvature constructed from it, an identity obeyed by that curvature, and a dynamical equation coupling the curvature to a source. We begin with the four-vector formulation of electromagnetism and the electromagnetic field tensor $F_{\mu\nu}$, including its explicit matrix representation and that of its dual. We then introduce the wedge product, the exterior derivative $\mathbf{d}$, and the Hodge star $\star$, and show that the four Maxwell equations collapse into the two statements $\mathbf{d}F = 0$ and $\mathbf{d} \star F = \mu_0 \star J$, with charge conservation following from $\mathbf{d}^2 = 0$. The same machinery is then carried over to gravitation, where the $U(1)$ connection is replaced by a local Lorentz connection and the curvature two-form acquires a quadratic self-interaction term. The Bianchi identity forces the Einstein tensor to be divergence-free, and Einstein's equation follows once matter is included. We close by emphasizing both the structural isomorphism and the essential physical differences between the two theories.

Keywords: Maxwell's equations, differential forms, $\mathbf{d}^2 = 0$, Bianchi identity, Einstein equation

Usage: Course paper for Seminar on Electrodynamics.

I. INTRODUCTION

Electromagnetism and gravitation occupy rather different positions in a typical undergraduate curriculum. Maxwell's equations are introduced as four vector equations for the electric and magnetic fields $\boldsymbol{E}$ and $\boldsymbol{B}$, while general relativity is presented through the metric tensor $g_{\mu\nu}$, Christoffel symbols, and the Riemann curvature tensor. At first sight the two subjects appear to share little beyond the word "field." Historically, however, the connection is deep: Einstein's 1905 paper on special relativity is explicitly motivated by the electrodynamics of moving bodies, and the Lorentz covariance of Maxwell's equations was the primary clue that the Galilean notion of space and time had to be abandoned [1].

A second, less widely appreciated link between the two theories emerges once electrodynamics is rewritten geometrically. The language of differential forms, developed by Élie Cartan in the early twentieth century and later absorbed into modern differential geometry, has long been recognized as the natural setting for electromagnetism [6, 7]. What is often left implicit in undergraduate treatments is that this same language also discloses the logical skeleton of general relativity. Both theories involve a *connection*, a *curvature* built from that connection, an *identity* satisfied by the curvature, and a *dynamical equation* that couples the curvature to a source. In symbols, the parallel is

$$A \longrightarrow F = \mathbf{d}A \qquad \text{and} \qquad \omega \longrightarrow \mathcal{R} = \mathbf{d}\omega + \omega \wedge \omega.$$

The left-hand side is electrodynamics; the right-hand side is gravitation. Our goal is to make this correspondence precise and to understand where it breaks down.

The organization of the essay mirrors the logic of the correspondence itself. Section II reviews relativistic electrodynamics and introduces the field tensor and its dual. Section III develops differential forms and the exterior derivative, leading to the source-free equation. Section IV adds the Hodge star and the sourced equation. Sections V and VI carry the construction over to gravitation, and Sec. VII explains how electrodynamics is modified in curved spacetime. Section VIII collects the structural isomorphism and its limitations.

We follow closely the presentation of Carroll's *Spacetime and Geometry* [2], in particular Secs. 1.8, 2.9–2.10, 3.2–3.7 and 4.2. Throughout we adopt the metric signature $(-,+,+,+)$ and, unless otherwise indicated, units in which $c = 1$. Greek indices run over $0, 1, 2, 3$ and Latin indices over $1, 2, 3$; repeated indices are summed.

II. RELATIVISTIC ELECTRODYNAMICS

A. Maxwell's equations and the speed-of-light problem

In SI units Maxwell's equations read

$$\nabla \cdot \boldsymbol{E} = \rho/\varepsilon_0, \qquad \nabla \times \boldsymbol{E} = -\partial_t \boldsymbol{B}, \tag{1}$$

$$\nabla \cdot \boldsymbol{B} = 0, \qquad \nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \, \partial_t \boldsymbol{E}. \tag{2}$$

In vacuum these equations imply a wave equation whose propagation speed, $1/\sqrt{\mu_0 \varepsilon_0} \equiv c$, is a *constant of the theory* and carries no reference to any frame. Galilean

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kinematics, by contrast, would predict that the measured speed shifts to $c \mp v$ in a frame moving with velocity v. Something must therefore give way: either electrodynamics, or Newtonian space and time. Einstein's two postulates—the principle of relativity and the constancy of the speed of light—select the second option and lead to the Lorentz transformation

$$x' = \gamma(x - vt), \qquad t' = \gamma\Big(t - \frac{vx}{c^2}\Big)\,, \\ \gamma = (1 - v^2/c^2)^{-1/2}. \tag{3}$$

The invariant of this transformation is the spacetime interval $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}\boldsymbol{x}^2$, which we write as $\mathrm{d}s^2 = \eta_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu$ with $\eta_{\mu\nu} = \mathrm{diag}(-1,+1,+1,+1)$. Setting $c = 1$ then makes time and space dimensionally identical and turns c into a mere unit-conversion factor.

B. The electromagnetic field tensor

The scalar and vector potentials combine into a four-vector $A^\mu = (\varphi, \boldsymbol{A})$, and the charge and current densities into $J^\mu = (\rho, \boldsymbol{J})$. From A_μ we construct the electromagnetic field tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \equiv 2\,\partial_{[\mu}A_{\nu]}, \tag{4}$$

where square brackets denote full antisymmetrization with weight $1/k!$,

$$T_{[\mu\nu]} \equiv \tfrac{1}{2!}\,(T_{\mu\nu} - T_{\nu\mu})\,. \tag{5}$$

Because $F_{\mu\nu} = -F_{\nu\mu}$, the tensor has only $4\times 3/2 = 6$ independent components—precisely the three components of $\boldsymbol{E}$ plus the three of $\boldsymbol{B}$. Explicitly, with $F_{0i} = -E_i$ and $F_{ij} = \varepsilon_{ijk}B^k$,

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \tag{6}$$

The contravariant tensor $F^{\mu\nu}$ differs only by an overall sign flip of the time row and column, since $\eta^{00}\eta^{ii} = -1$ whereas $\eta^{ii}\eta^{jj} = +1$.

It is also convenient to introduce the *dual* field tensor

$$(\star F)_{\mu\nu} \equiv \tfrac{1}{2}\,\varepsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}, \qquad \varepsilon_{0123} = +1, \tag{7}$$

whose matrix is obtained from Eq. (6) by the replacement $\boldsymbol{E} \to -\boldsymbol{B}$, $\boldsymbol{B} \to \boldsymbol{E}$:

$$(\star F)_{\mu\nu} = \begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & E_z & -E_y \\ -B_y & -E_z & 0 & E_x \\ -B_z & E_y & -E_x & 0 \end{pmatrix}. \tag{8}$$

In this language the four Maxwell equations compress into two tensor statements: the *sourced* equation

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu \tag{9}$$

(the Gauss and Ampère–Maxwell laws), and the *source-free* equation

$$\partial_{[\mu}F_{\nu\rho]} = 0 \iff \partial_\mu(\star F)^{\mu\nu} = 0 \tag{10}$$

(the magnetic Gauss law and Faraday's law). Note that the second equation is an *identity* once F is written in terms of a potential, whereas the first is genuinely dynamical. This distinction will reappear in gravitation.

C. Three motivations for a better language

Although Eqs. (9) and (10) are compact, they are unsatisfying in three respects. First, one must keep track of upper and lower indices, the summation convention, and antisymmetry simultaneously. Second, the antisymmetry of $F_{\mu\nu}$ has to be imposed by hand, even though it is clearly an intrinsic feature of the object. Third, the Levi-Civita symbol must be written out explicitly every time a dual is taken. The formalism of differential forms resolves all three issues at once: antisymmetry becomes the wedge product $\wedge$, differentiation becomes the exterior derivative $\mathbf{d}$, and duality becomes the Hodge star $\star$.

III. DIFFERENTIAL FORMS AND THE EXTERIOR DERIVATIVE

A. One-forms and the wedge product

A one-form is a $(0,1)$ tensor, i.e. a dual vector. With respect to the dual basis $\{\mathrm{d}x^\mu\}$, defined by $\langle \mathrm{d}x^\mu, \partial_\nu\rangle = \delta^\mu{}_\nu$, any one-form can be written as $A = A_\mu \mathrm{d}x^\mu$. The components A_μ depend on the coordinate system, but A itself does not. The electromagnetic potential is naturally such an object.

The wedge product of two one-forms is defined by antisymmetry,

$$\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu = -\mathrm{d}x^\nu \wedge \mathrm{d}x^\mu, \qquad \mathrm{d}x^\mu \wedge \mathrm{d}x^\mu = 0, \tag{11}$$

and extended bilinearly. For two one-forms ω and η this gives

$$\omega \wedge \eta = \tfrac{1}{2}\,(\omega_\mu\eta_\nu - \omega_\nu\eta_\mu)\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu, \tag{12}$$

so that antisymmetrization is performed automatically, without any $[\ ]$ notation. A p-form is a completely antisymmetric $(0,p)$ tensor; in four dimensions the number of independent components is $\binom{4}{p} = 1, 4, 6, 4, 1$ for $p = 0, 1, 2, 3, 4$. The six components of a two-form are exactly what is needed to house $\boldsymbol{E}$ and $\boldsymbol{B}$ together. Crucially, p-forms can be defined using only the smooth structure of the manifold—no metric is required.

B. The exterior derivative and $\mathbf{d}^2 = 0$

The exterior derivative maps a p-form to a $(p+1)$-form via

$$\mathbf{d}(f\,\mathrm{d}x^\mu \wedge \cdots) = \mathrm{d}f \wedge \mathrm{d}x^\mu \wedge \cdots . \tag{13}$$

Its defining algebraic property is

$$\mathbf{d}^2 = 0, \tag{14}$$

which follows immediately because $\partial_\mu\partial_\nu f$ is symmetric while $\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$ is antisymmetric. Like the wedge product, $\mathbf{d}$ requires no metric and no preferred coordinate system; it is a purely topological operation.

Applied to the potential one-form, the exterior derivative reproduces the field tensor:

$$\begin{aligned}\mathbf{d}A &= \partial_\nu A_\mu\,\mathrm{d}x^\nu \wedge \mathrm{d}x^\mu \\ &= \tfrac{1}{2}\left(\partial_\mu A_\nu - \partial_\nu A_\mu\right)\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu = \tfrac{1}{2}F_{\mu\nu}\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu,\end{aligned} \tag{15}$$

so that $F = \mathbf{d}A$. A gauge transformation becomes $A \to A + \mathbf{d}\lambda$, and Eq. (14) then guarantees

$$F \;\longrightarrow\; F + \mathbf{d}^2\lambda = F. \tag{16}$$

Gauge invariance is therefore not an additional assumption but a direct consequence of $\mathbf{d}^2 = 0$.

C. The source-free equation

The source-free Maxwell equation follows at once:

$$\mathbf{d}F = \mathbf{d}^2 A = 0. \tag{17}$$

Expanding in components yields $\nabla \cdot \boldsymbol{B} = 0$ and $\nabla \times \boldsymbol{E} + \partial_t \boldsymbol{B} = 0$, as required. Conversely, the Poincaré lemma states that in a simply connected region $\mathbf{d}F = 0$ implies the existence of a potential A with $F = \mathbf{d}A$. The existence of a potential and the source-free equation are thus two faces of the same statement. If magnetic monopoles existed, one would instead have $\mathbf{d}F \neq 0$, and A could not be defined globally—a genuinely topological obstruction. We emphasize again that Eq. (17) contains no metric: it will survive unchanged in curved spacetime.

IV. HODGE DUALITY AND THE SOURCED EQUATION

A. The Hodge star

On an n-dimensional manifold equipped with a metric, the Hodge star maps a p-form to an $(n-p)$-form:

$$(\star A)_{\mu_1\cdots\mu_{n-p}} = \frac{1}{p!}\,\varepsilon^{\nu_1\cdots\nu_p}{}_{\mu_1\cdots\mu_{n-p}} A_{\nu_1\cdots\nu_p}. \tag{18}$$

In four dimensions it interchanges one-forms with three-forms and maps two-forms to two-forms. For two-forms in Lorentzian signature one finds $\star\star F = -F$. In three-dimensional Euclidean space the Hodge star reproduces a familiar object: $\star(U \wedge V)_i = \varepsilon_{ijk}U^jV^k$ is precisely the cross product $\boldsymbol{U} \times \boldsymbol{V}$. The Hodge dual is therefore not a new operation but the generalization of the cross product to arbitrary dimension.

The essential difference from $\mathbf{d}$ is that $\star$ *depends on the metric*: the Levi-Civita tensor carries a factor $\sqrt{|g|}$. While $\mathbf{d}$ is topological, $\star$ is geometrical. This asymmetry is the doorway to gravity.

B. The sourced Maxwell equation

With $J = J_\mu \mathrm{d}x^\mu$ the current one-form, $\star J$ is a three-form, and the sourced Maxwell equation takes the form

$$\mathbf{d} \star F = \mu_0 \star J. \tag{19}$$

Both sides are three-forms, so the equation is type-consistent. Expanding in components returns Eq. (9), i.e. the Gauss and Ampère–Maxwell laws. Charge conservation is now automatic: applying $\mathbf{d}$ once more and using $\mathbf{d}^2 = 0$ gives

$$0 = \mathbf{d}^2 \star F = \mu_0\,\mathbf{d} \star J \quad\Longleftrightarrow\quad \partial_\mu J^\mu = 0. \tag{20}$$

Conservation of charge is not an independent postulate but an integrability condition of the field equation. Finally, Stokes' theorem $\int_{\mathcal{M}} \mathbf{d}\omega = \int_{\partial\mathcal{M}} \omega$ converts Eqs. (17) and (19) into the familiar integral laws of Gauss, Faraday and Ampère, differing only in the dimension of the integration region.

V. FROM $U(1)$ CONNECTIONS TO GRAVITATIONAL CONNECTIONS

General relativity replaces the fixed Minkowski metric by a dynamical field $g_{\mu\nu}(x)$. The equivalence principle suggests that freely falling frames are locally inertial, so spacetime must be allowed to curve. Once $\eta_{\mu\nu} \to g_{\mu\nu}(x)$, ordinary partial derivatives of vectors are no longer tensors, and a connection is needed to compare vectors at neighbouring points.

The most transparent way to see the analogy with electrodynamics is through the frame (vielbein) formalism. One introduces an orthonormal frame $e^a = e^a{}_\mu \mathrm{d}x^\mu$ and a local Lorentz connection $\omega^a{}_b = \omega^a{}_{b\mu}\mathrm{d}x^\mu$, with $g_{\mu\nu} = \eta_{ab}e^a{}_\mu e^b{}_\nu$. The electromagnetic connection A and the gravitational connection $\omega^a{}_b$ are objects of the same kind; what changes is the gauge group, from the abelian $U(1)$ to the local Lorentz group $SO(1,3)$. Demanding vanishing torsion and metric compatibility,

$$T^a = \mathbf{d}e^a + \omega^a{}_b \wedge e^b = 0, \qquad De^a = 0, \tag{21}$$

fixes $\omega^a{}_b$ completely in terms of the frame, just as the Levi-Civita connection is fixed by the metric.

The curvature two-form is then

$$\mathcal{R}^a{}_b = \mathbf{d}\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b = \tfrac{1}{2} R^a{}_{b\mu\nu}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu, \qquad (22)$$

whose components $R^a{}_{b\mu\nu}$ are the Riemann tensor in the local frame. Geometrically, parallel transporting a frame around a closed loop returns it rotated by a local Lorentz transformation; the infinitesimal generator of that rotation is $\mathcal{R}^a{}_b$. This holonomy is the precise meaning of curvature.

Comparing Eqs. (22) with $F = \mathbf{d}A$ reveals the first essential difference: the electromagnetic curvature is *linear* in the connection, whereas the gravitational curvature contains the quadratic term $\omega \wedge \omega$. The non-abelian nature of the gauge group makes gravity self-interacting.

VI. BIANCHI IDENTITY AND EINSTEIN'S EQUATION

The curvature two-form obeys the Bianchi identity

$$D\mathcal{R}^a{}_b = \mathbf{d}\mathcal{R}^a{}_b + \omega^a{}_c \wedge \mathcal{R}^c{}_b - \mathcal{R}^a{}_c \wedge \omega^c{}_b = 0, \qquad (23)$$

which in tensor language reads $\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0$. Contracting twice yields the contracted Bianchi identity

$$\nabla^\mu \big(R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} \big) = 0, \qquad (24)$$

where $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$ is the Ricci tensor and $R = g^{\mu\nu} R_{\mu\nu}$ the Ricci scalar. Defining the Einstein tensor $G_{\mu\nu} \equiv R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu}$, we have automatically $\nabla^\mu G_{\mu\nu} = 0$.

This is exactly what is needed for a consistent coupling to matter. The stress-energy tensor satisfies $\nabla^\mu T_{\mu\nu} = 0$ as a consequence of diffeomorphism invariance of the matter action. If geometry is to be equated to matter, the geometric side must be identically divergence-free, and the Bianchi identity guarantees that $G_{\mu\nu}$ is the simplest such combination. The field equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu} \qquad (25)$$

then follows, with the constant fixed by the Newtonian limit. The same equation can be derived from the Hilbert action $S = \frac{c^4}{16\pi G} \int R\sqrt{-g}\, \mathrm{d}^4x + S_{\mathrm{M}}$ by variation with respect to $g^{\mu\nu}$.

VII. ELECTRODYNAMICS IN CURVED SPACETIME

The formalism developed above also answers a practical question: how should Maxwell's equations be written on a curved background? The answer is remarkably economical. Because the source-free equation $\mathbf{d}F = 0$ involves only the exterior derivative, it is completely insensitive to the metric and therefore survives unchanged. The sourced equation, by contrast, contains the Hodge star, which depends on the metric through $\sqrt{|g|}$; the only modification required is to replace $\eta_{\mu\nu}$ by $g_{\mu\nu}$ inside $\star$. In coordinate form this amounts to the usual minimal-coupling prescription $\partial_\mu \to \nabla_\mu$ together with the invariant volume element $\sqrt{-g}\, \mathrm{d}^4x$. Gravity thus enters electrodynamics through exactly one channel, and no new physical assumption is needed. This observation also clarifies why the two Maxwell equations play such different roles in the structure of the theory: one is a topological constraint, the other a genuinely metric-dependent dynamical law.

VIII. STRUCTURAL ISOMORPHISM AND PHYSICAL DIFFERENCES

The correspondence can be summarized in a single table:

	Electromagnetism	Gravitation
Gauge group	$U(1)$	local $SO(1,3)$
Connection	$A = A_\mu \mathrm{d}x^\mu$	$\omega^a{}_b$
Curvature	$F = \mathbf{d}A$	$\mathcal{R} = \mathbf{d}\omega + \omega \wedge \omega$
Identity	$\mathbf{d}F = 0$	$D\mathcal{R} = 0$
Dynamics	$\mathbf{d} \star F = \mu_0 \star J$	$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$
Conservation	$\partial_\mu J^\mu = 0$	$\nabla^\mu T_{\mu\nu} = 0$

Both theories follow the pattern *connection → curvature → identity + field equation*, and both conservation laws are integrability conditions rather than independent assumptions.

Yet the differences are just as instructive. First, the metric $g_{\mu\nu}$ is not a gauge potential on the same footing as A_μ; the correct correspondence is $A_\mu \leftrightarrow \omega^a{}_{b\mu}$. Second, gravity is nonlinear both in its curvature and in its field equation, because the gravitational field itself carries energy and therefore gravitates. Third, the Hodge star makes the sourced Maxwell equation metric-dependent, whereas $\mathbf{d}F = 0$ is metric-free; in curved spacetime only the latter survives unchanged, and the former is modified simply by replacing η with g inside $\star$. Finally, electric charge comes in both signs while mass-energy does not, which is why gravity can collapse into black holes but electromagnetism cannot.

IX. CONCLUSION

Rewriting Maxwell's equations in the language of differential forms is far more than a notational trick. The wedge product absorbs antisymmetry, the exterior derivative absorbs differentiation, and the identity $\mathbf{d}^2 = 0$ simultaneously delivers gauge invariance, the source-free Maxwell equation, and charge conservation. The Hodge star then supplies the sourced equation, at the price of introducing metric dependence. Carrying the same machinery over to gravitation, with the gauge group

promoted from $U(1)$ to the local Lorentz group, leads directly to the curvature two-form, the Bianchi identity, and finally Einstein's equation. Differential forms thus provide a common grammar for the two theories, while the gauge group, the nonlinearity, and the physical interpretation remain specific to each.

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